

GE Healthcare

Discovery MR450 & MR750
Field Replaceable Units (FRU)

Technical Publication

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



5500105
Rev 5

**COVER
BOTTOM
CROP LINE**

**COVER
BOTTOM
CROP LINE**

**COVER
FOLD
LINE** | **(2) CENTER STAPLES
FOR BINDING**

**COVER
RIGHT
CROP LINE**

**5/16" RADIUS
CORNER TRIM
TYPICAL**

**5/16" RADIUS
CORNER TRIM
TYPICAL**

**COVER
LEFT
CROP LINE**

SEND COMMENTS ABOUT THIS MANUAL TO:

GE MEDICAL SYSTEMS
MANAGER, SERVICE ENGINEERING
W-808
3200 N. GRANDVIEW BLVD.
WAUKESHA, WISCONSIN 53188
USA

This manual belongs to:

INTRODUCTION TO FIELD REPLACEABLE UNIT (FRU)

This direction is a guide to the FRU's.

Parts designated as a FRU are critical to maintain the safety and/or functionality of the system. FRU's are available on a Priority 1 basis from Service Parts Operation.

As a reference aid, the FRU's are listed under an upper-level part number where they are used. In most cases the upper-level part number is not a FRU. In some cases the parts listed under an upper-level assembly may have been nested in intermediate sub-assemblies which are not shown.

Where possible, FRU's which are part of Option Catalogs are shown under the area where they would be installed, rather than being listed in the Coils or Options Sections (SEE TABLE OF CONTENTS).

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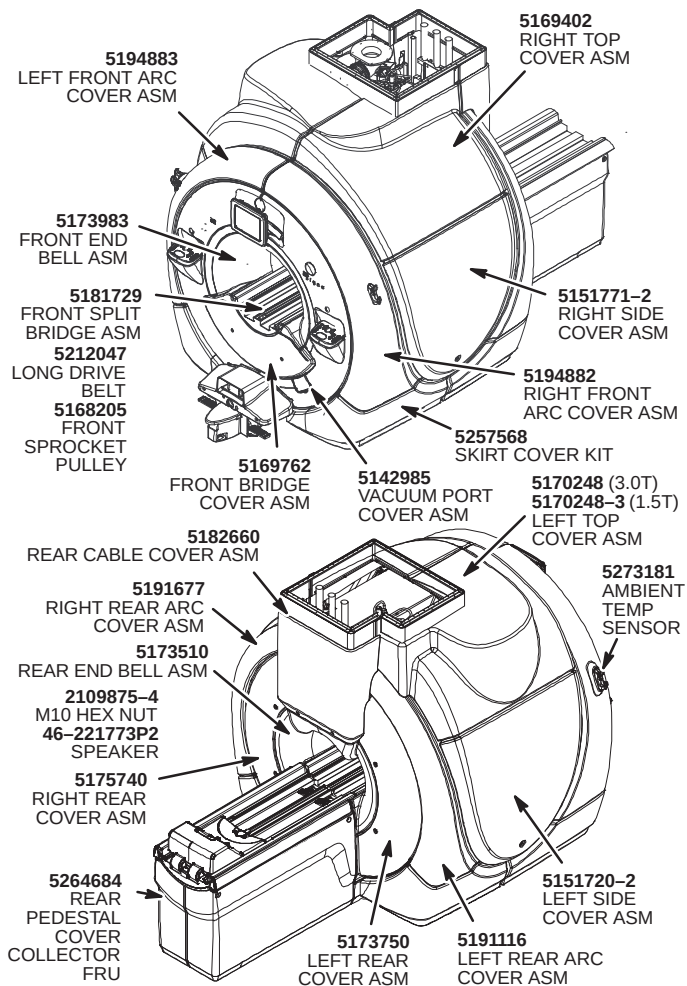
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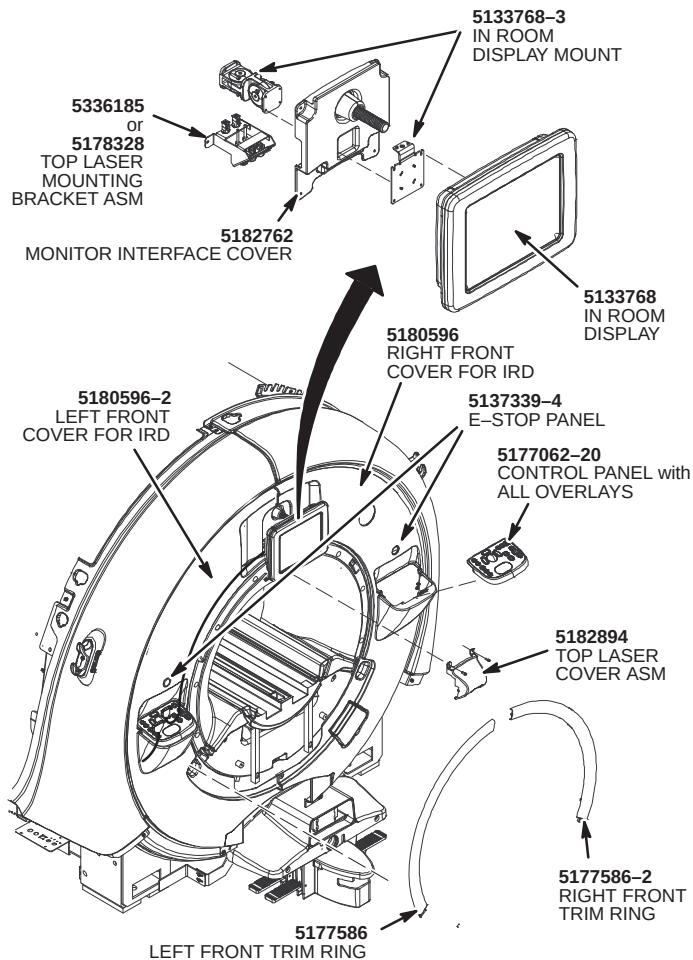
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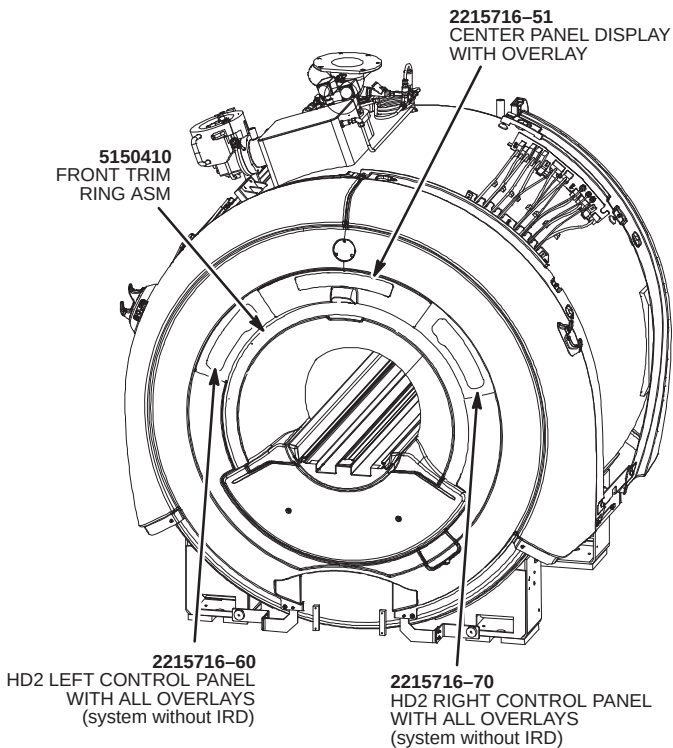
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COMMON ENCLOSURE COMPONENTS**MAGNET ENCLOSURE**

ENCLOSURE COMPONENTS for IN-ROOM DISPLAY (IRD)

**ENCLOSURE COMPONENTS for SEVEN SGMENT DISPLAY
(without IRD)**



MAGNET ENCLOSURE

ENCLOSURE ELECTRONIC COMPONENTS**COLD HEAD FRU's**

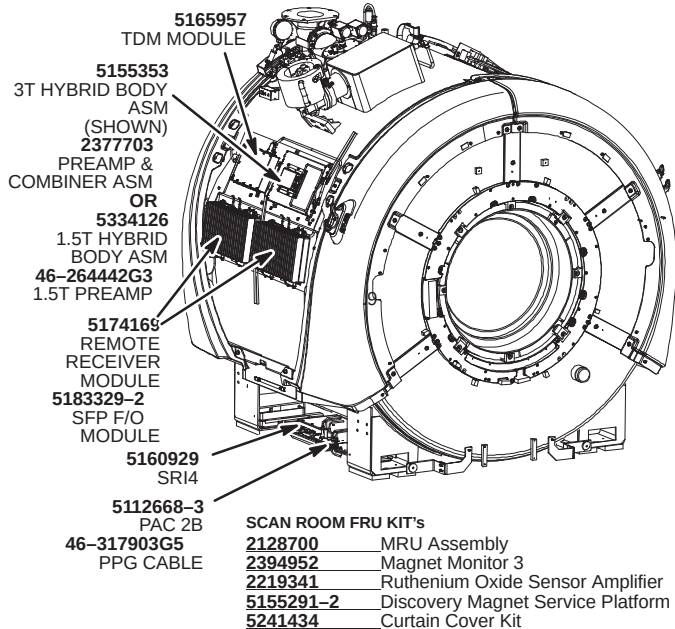
2154502-2	Run 621, Supply Gas Line, 65.6 ft or 20M long
2154505-2	Run 622, Return Gas Line, 65.6 ft or 20M long
2155316-2	Run 623, Compressor to Pen Panel Cable, 49.21 ft or 15M long
2218292	Run 624, Cold Head Control Cable
2214062	Run 832, LCC Sleeve and Cold Head Cable
2259837	LCC Cold Head Control Cable Asm

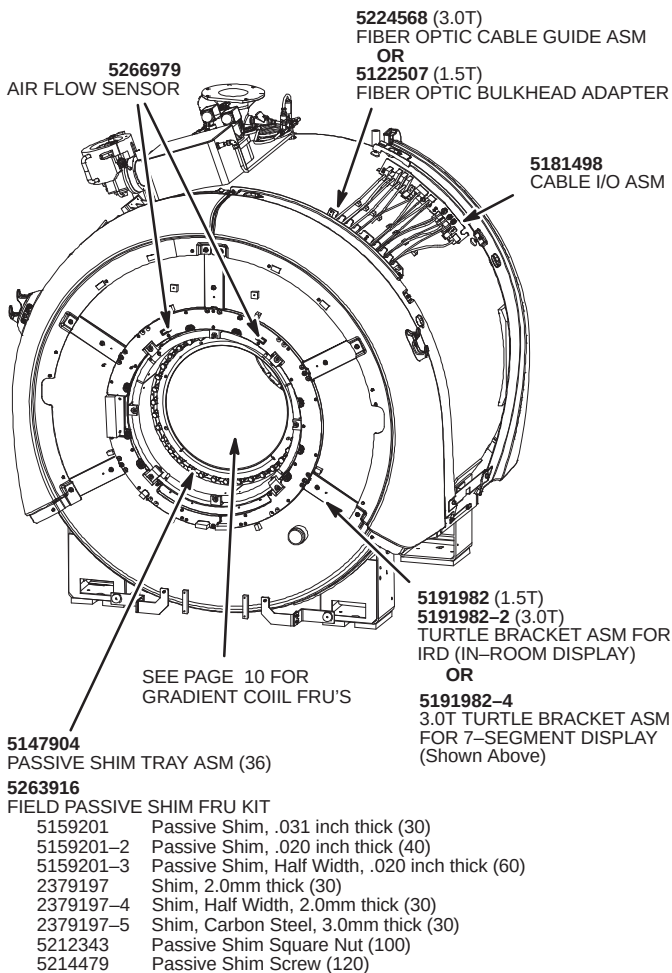
MAGNET FRU's

46-260852G3	Emergency Evacuation Release Tool
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CABLE FRU's

5304950	RF Cable FRU's for Body Hybrid to Body Coil (Contains Runs M1322 and M1323)
5183293	Microphone Cable, Run M3345
5137339-6	E-Stop Cable
5180907	Run 3350, MAG-SRI-J10 to MAG-BORE, Bore Temp,



ENCLOSURE AND SHIM COMPONENTS**MAGNET ENCLOSURE**

MR450 1.5T GRADIENT AND RF BODY COIL**2208999-14**

1.5T RF BODY COIL Includes
ORANGE & BLACK FRU CASE
AND SKID

2144093-12

GRADIENT COIL with HOS
Includes GRAY COIL CART
WITH CRADLE

5316997**COOLANT MANIFOLD COLLECTOR**

FRU contains:

- 5268005, Coolant Supply Manifold – BLU
- 5268152, Coolant Supply Manifold – BLK
- 46-208765P2Hose Clamps (25)
- 46-294167P91 Electrical Tape

5307840

BUSBAR FRU ASM contains:

- 5305786, Leadboard
- 46-294167P96, Bushing (8)
- 5268439, Axial Adj. Shim (12)
- 5214524-6, M10 SS Washer (20)
- 2341885, M10 Brass Nut (22)
- 5308479, Installation Gage Tool

5197020

HOS CONNECTOR
TERMINATION CAP

5191537 (1.5T)

Rear Turtle Bracket Asm

5335389

1.5T GRADIENT MOUNTING FRU KIT contains:
(For mounting front and rear of coil)

- 5170015 Radial Positioning Block (2)
- 5170098 Axi-Radial Positioning Block (4)
- 5170312 Inward Mounting Wedge (5)
- 5170035 Outward Mounting Wedge (4)
- 5326293-4 M10 x 30mm Hex Head Cap Screw (15)
- 5146169 M10 x 24mm Washer (15)
- 46-294151P11 Locktite 271

MR750 3.0T GRADIENT AND RF BODY COIL**2208999-13**

3T RF BODY COIL Includes
ORANGE FRU CASE AND SKID

2144093-12

GRADIENT COIL with HOS
Includes GRAY COIL CART
WITH CRADLE

5316997

COOLANT MANIFOLD COLLECTOR

FRU contains:

5268005, Coolant Supply Manifold – BLU

5268152, Coolant Supply Manifold – BLK

46-208765P2Hose Clamps (25)

46-294167P91 Electrical Tape

5307840

BUSBAR FRU ASM contains:

5305786, Leadboard

46-294167P96, Bushing (8)

5268439, Axial Adj. Shim (12)

5214524-6, M10 SS Washer (20)

2341885, M10 Brass Nut (22)

5308479, Installation Gage Tool

5197020

HOS CONNECTOR
TERMINATION CAP

5191537-2 (3.0T)

Rear Turtle Bracket Asm

5307838

3.0T G3GRADIENT MOUNTING FRU KIT contains:
(For mounting front and rear of coil)

5155280 G3 Radial Positioning Block (2)

5154485 G3 Axi-Radial Positioning Block (4)

5166786 G3 Inward Mounting Wedge (5)

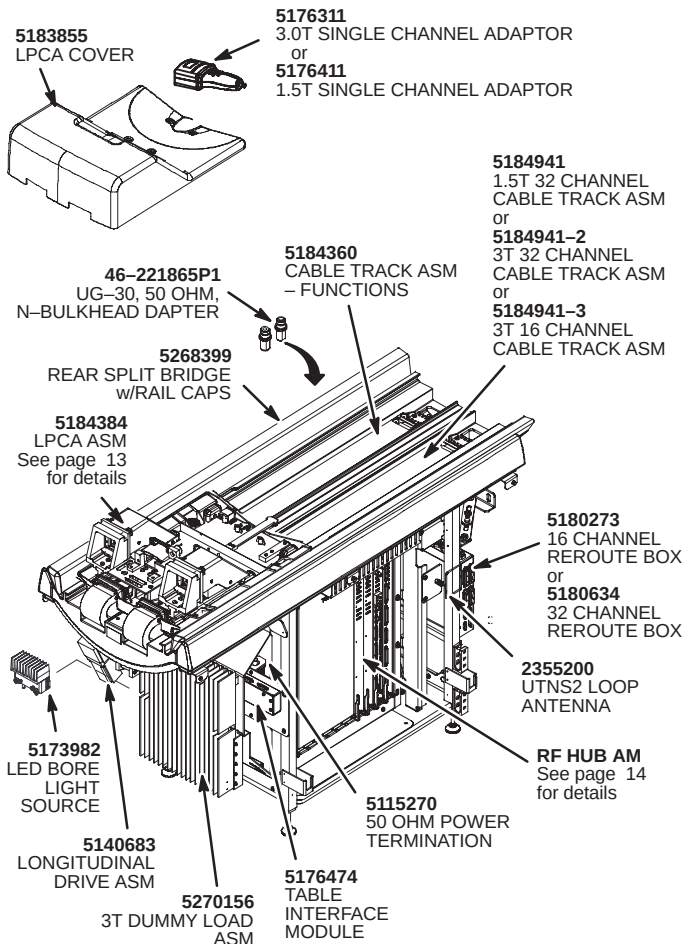
5166458 G3 Outward Mounting Wedge (4)

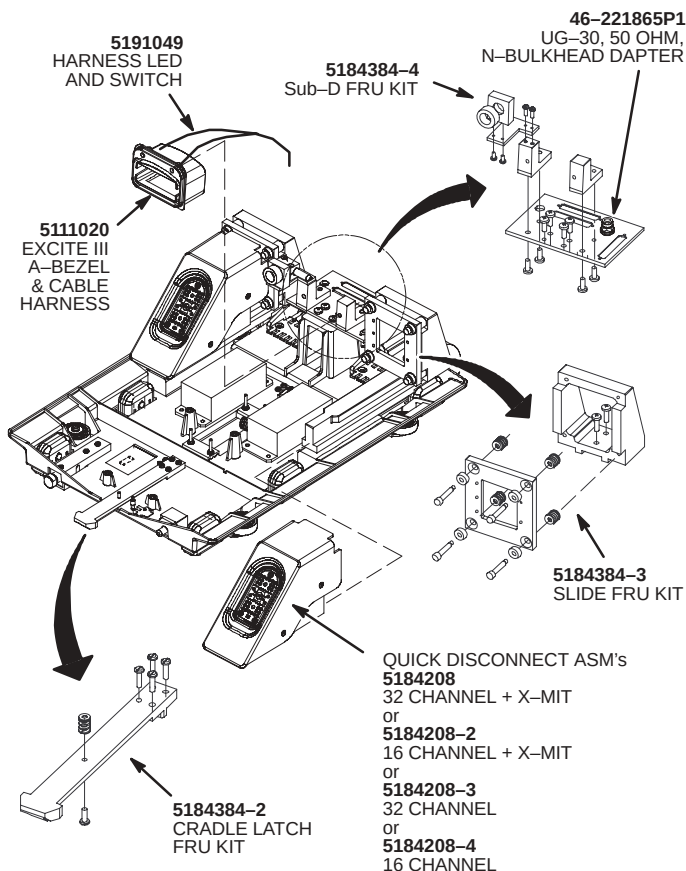
5110638-64 M10 x 30 Brass Bolt (15)

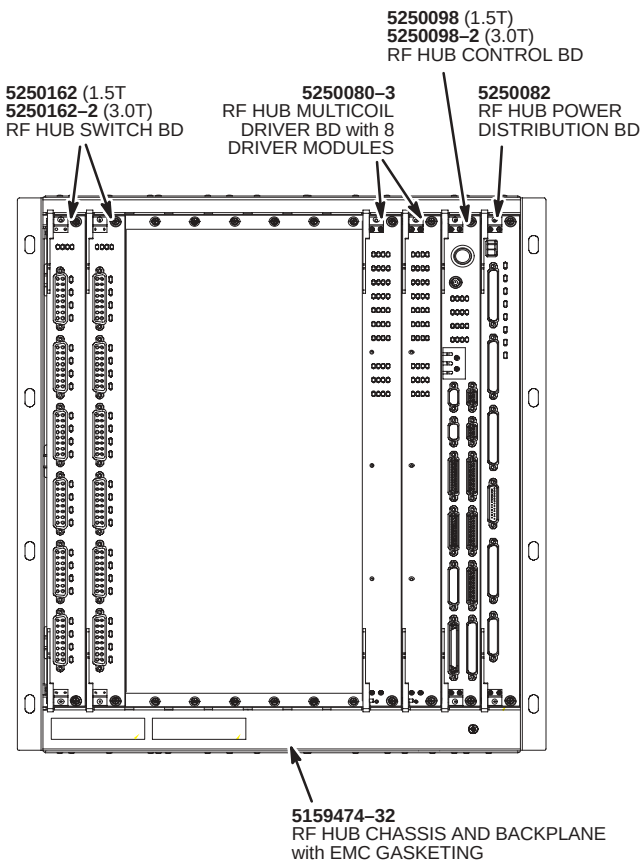
5146169 M10 x 24mm Washer (15)

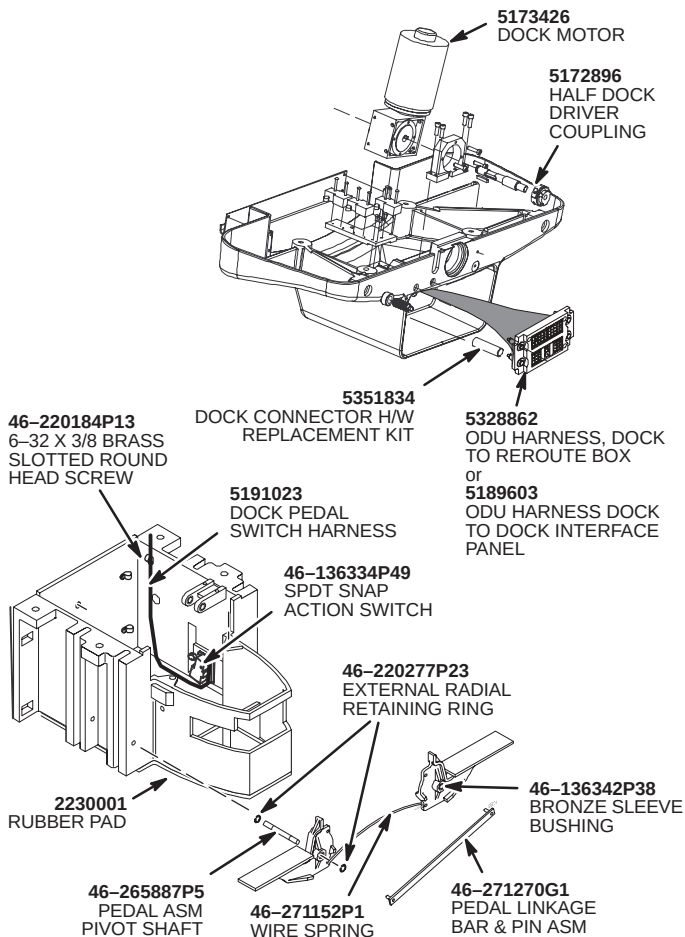
46-294151P11 Locktite 271

GRADIENT & RF BODY COILS

REAR PEDESTAL

LPCA**REAR PEDESTAL**

RF HUB

MAGNET DOCKING PARTS:

PATIENT TABLE

FRU part names, listed in numerical order, are found on pages 26 thru 31.

5368689 and 5367857 are for the oncology flat table only.

5368689
FLAT TABLE
INSERT

5367857
TABLE
FILLER

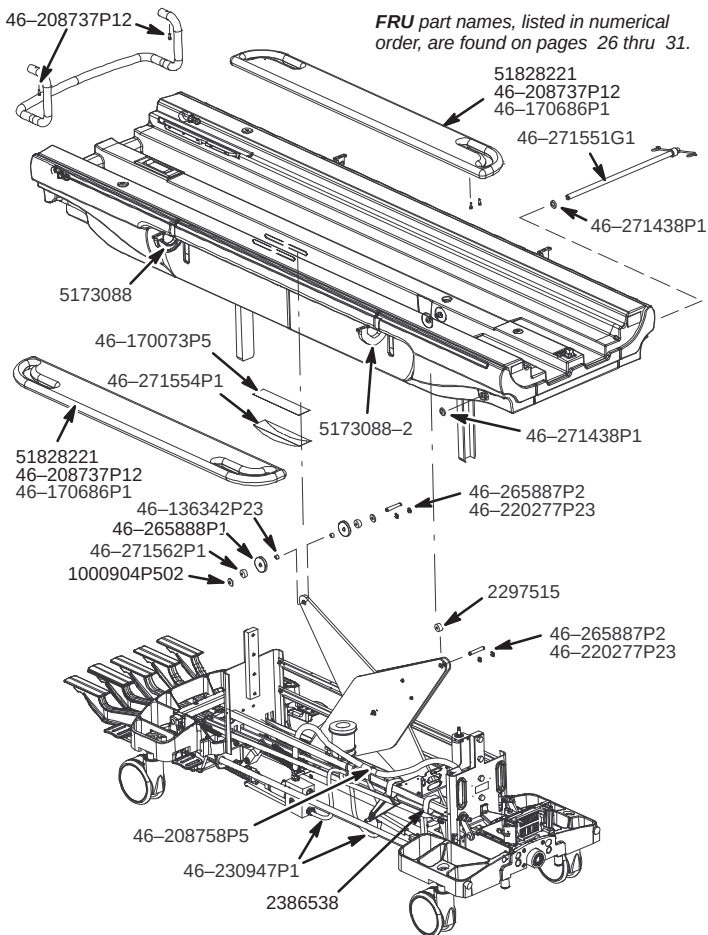
CRADLE DETAILS
See page 24

TABLE TOP DETAILS
See pages 18 and 25

5173360
TABLE BELLOWS
COVER

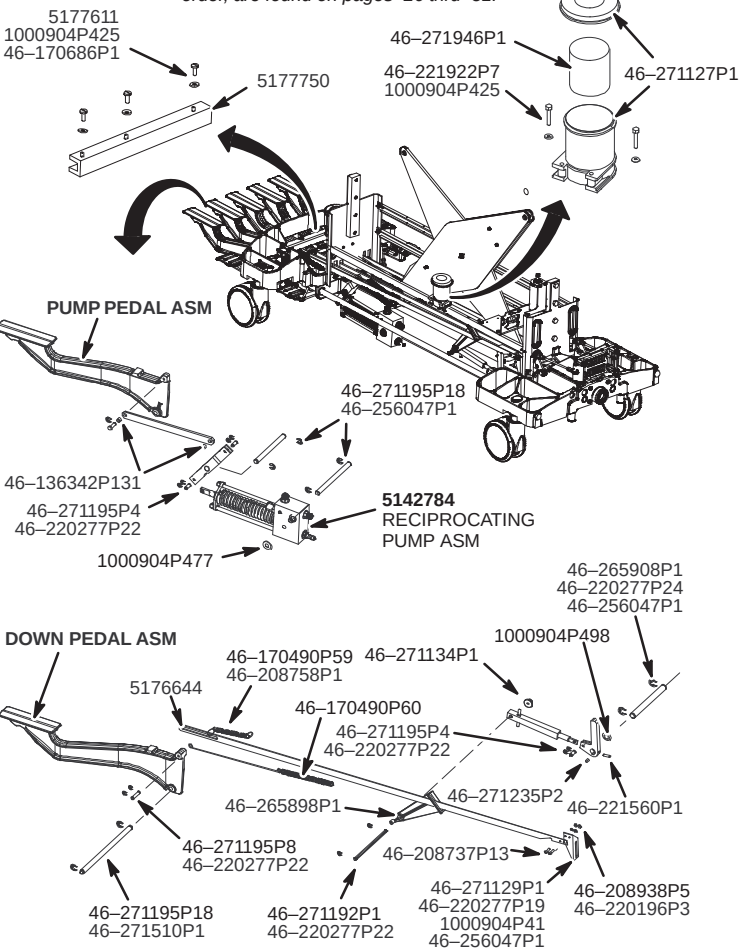
**TABLE LOWER
MECHANISMS DETAILS**
See pages 18 thru 24

PATIENT TABLE

PATIENT TABLE TOP & BOTTOM

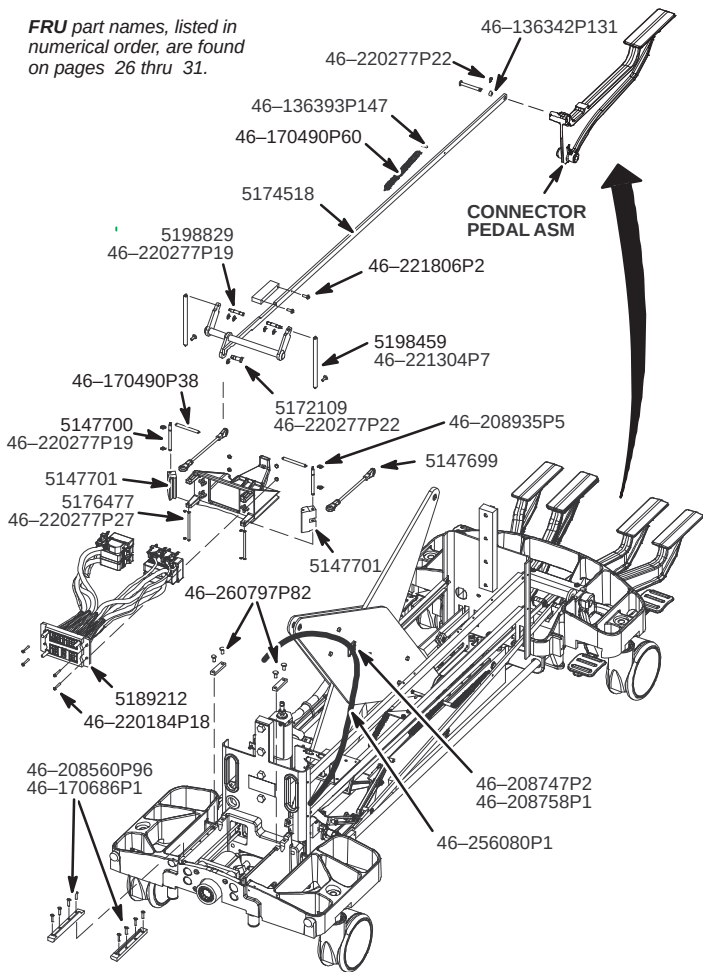
PATIENT TABLE PEDAL ASM'S

FRU part names, listed in numerical order, are found on pages 26 thru 31.

**PATIENT TABLE**

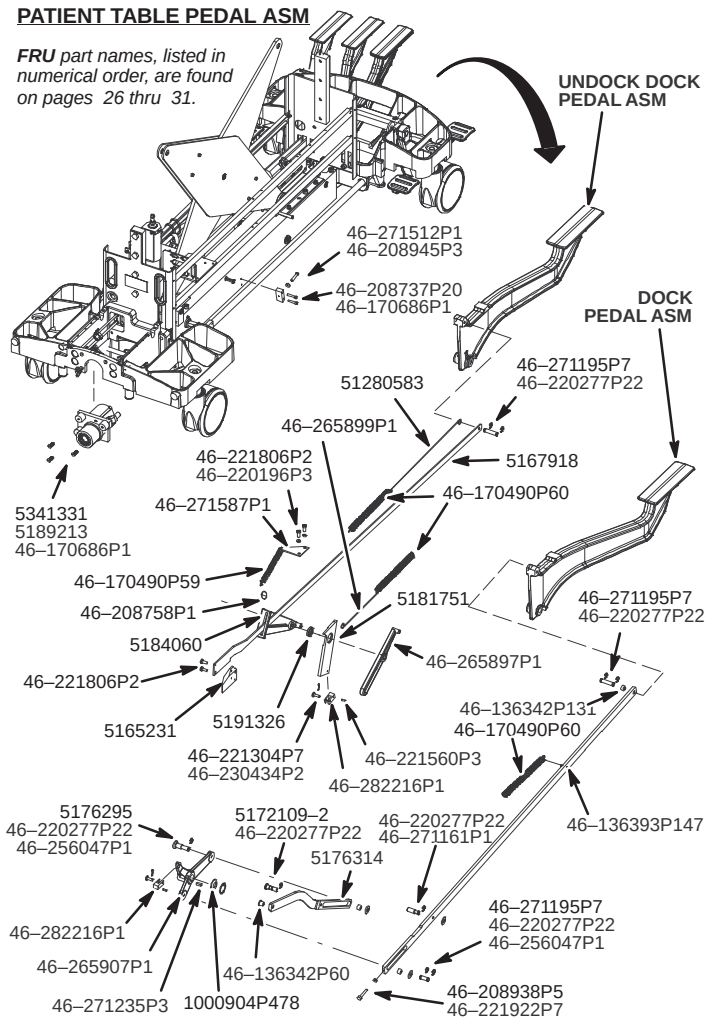
PATIENT TABLE PEDAL ASM'S

FRU part names, listed in numerical order, are found on pages 26 thru 31.



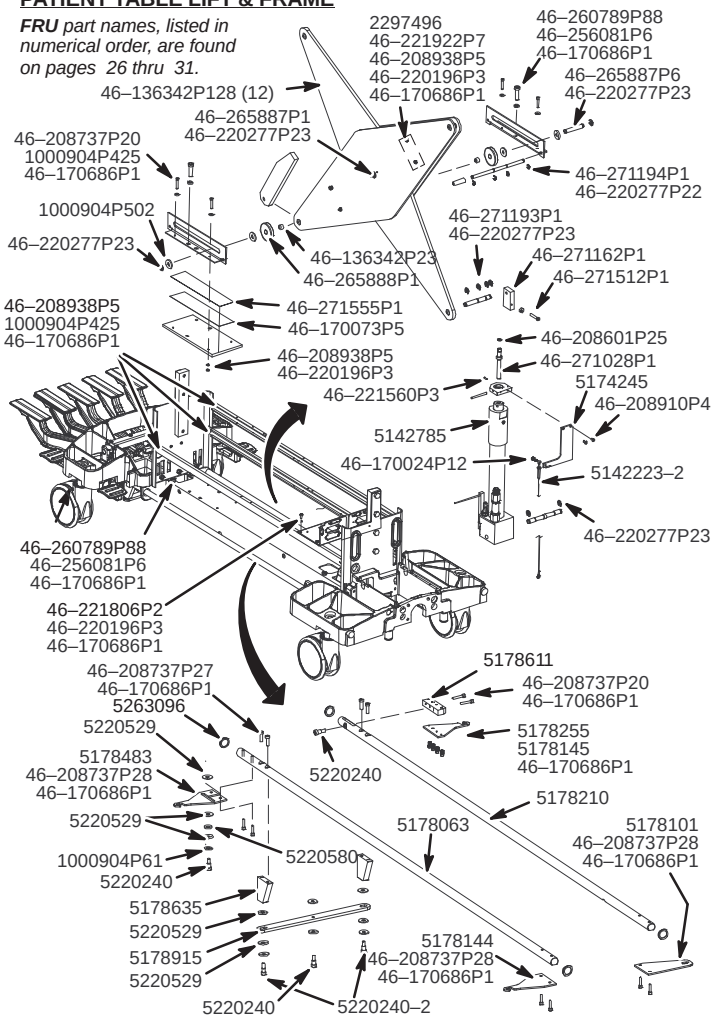
PATIENT TABLE PEDAL ASM

FRU part names, listed in numerical order, are found on pages 26 thru 31.

**PATIENT TABLE**

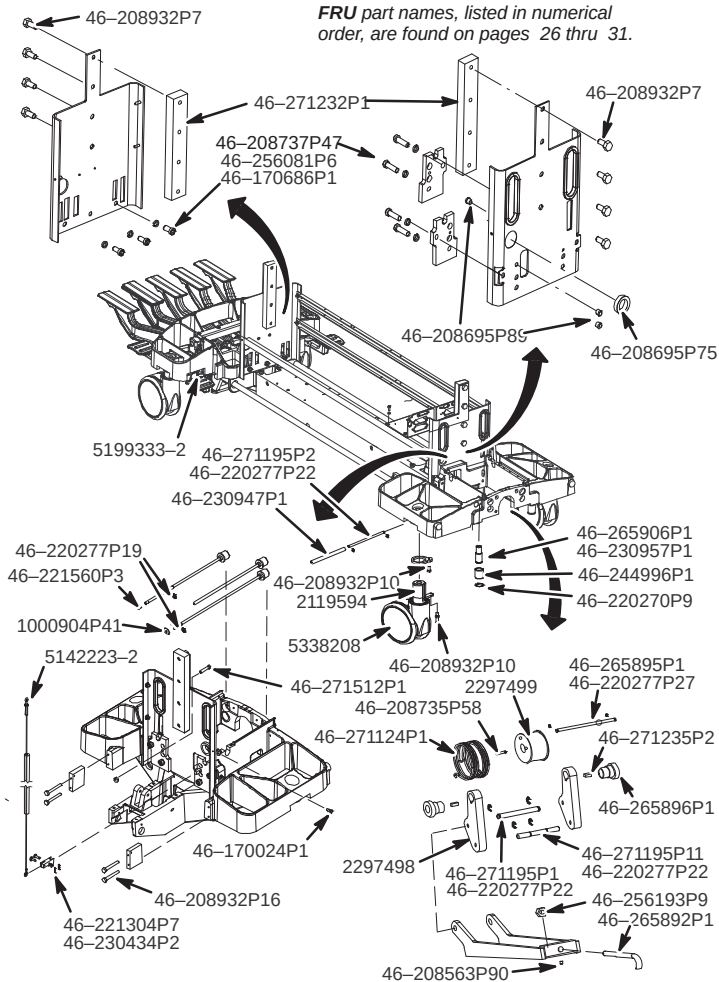
PATIENT TABLE LIFT & FRAME

FRU part names, listed in numerical order, are found on pages 26 thru 31.



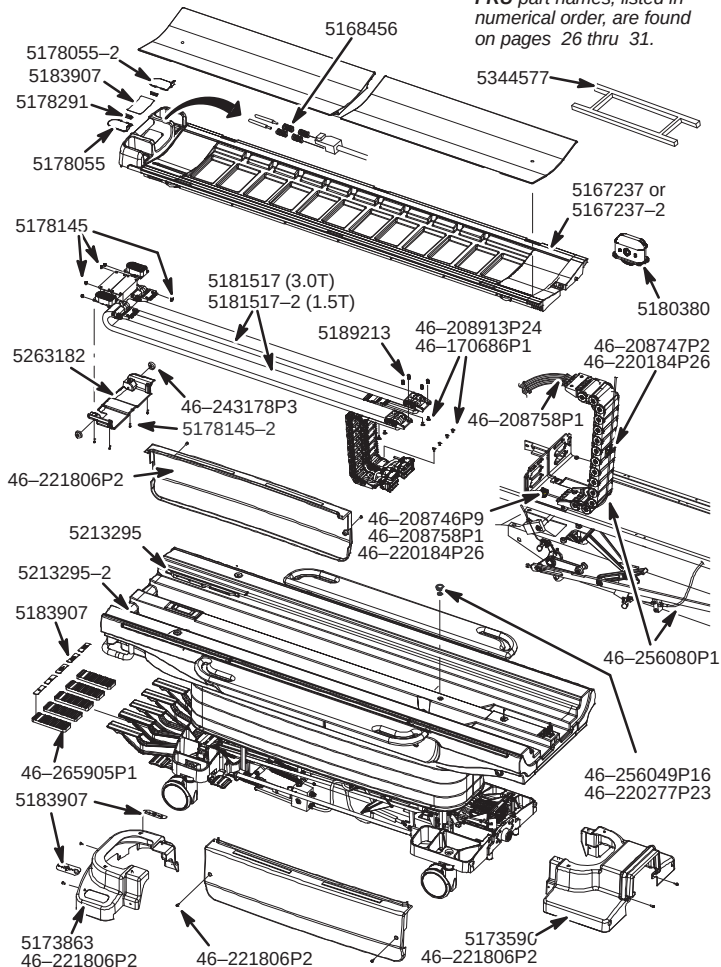
PATIENT TABLE DOCK AREA & CASTORS

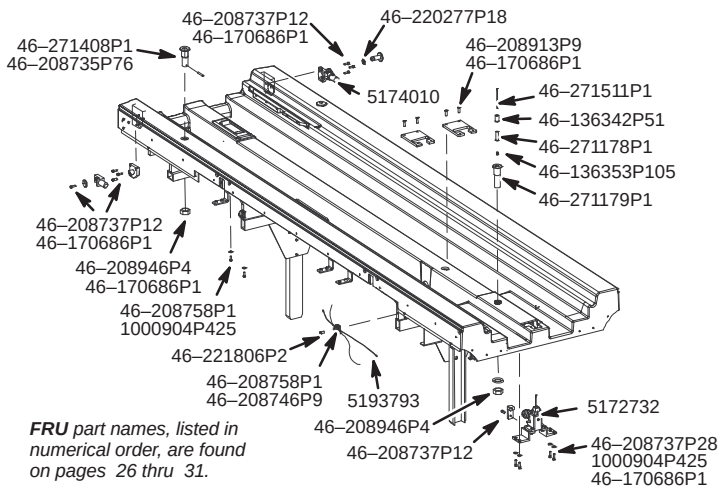
FRU part names, listed in numerical order, are found on pages 26 thru 31.

**PATIENT TABLE**

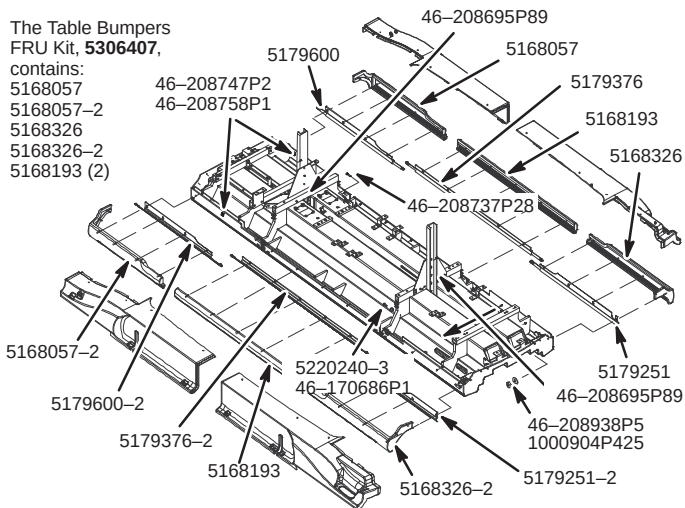
PATIENT TABLE CRADLE, CABLE TRACKS, & COVERS

FRU part names, listed in numerical order, are found on pages 26 thru 31.



PATIENT TABLE TOP FRU's

The Table Bumpers
FRU Kit, **5306407**,
contains:
5168057
5168057-2
5168326
5168326-2
5168193 (2)

**PATIENT TABLE**

PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
2119594	Anti-seize Lubricant; Permatex 133A
2297496	Spacer, Alum Bar 1/2~ X 3/4~ X 3.5~
2297499	Spring Drum 1.96~Dia, 1.2~W Aluminum
2297515	Round Spacer, 1 IN OD, .390 ID
2386538	Velcro Strap Type 889
5136198	Caster Wheel
5142223	Up Sensor Cable
5142223-2	Uplimit Dock
5142784	Reciprocating Pump Asm
5142785	Cylinder Asm
5147699	Rod Link Connector
5147700	Shaft Rocker
5147701	Rocker Clamp
5154883	Lateral Wheel Housing
5165231	Block Hook Undock
5167237	Cradle Asm
5167237-2	16 Channel Cradle Asm
5167918	Link Undock
5168057	Front LH Side Bumper
5168057-2	Front RH Side Bumper
5168193	Arm Board Bumper
5168326	Rear Left Hand Side Bumper
5168326-2	Rear Right Hand Side Bumper
5168456	Compression Spring 10mm, 4.5mm ID, 18mm L
5172109	Pin Undock Link
5172109-2	Pin Undock Link
5172732	Front Latch Asm
5172856	Cradle Stop Polyurethane
5173088	Rear Armboard Hinge Asm
5173088-2	Front Armboard Hinge Asm
5173360	Table Bellows Cover
5173590	Front Base Cover
5173863	Rear Base Cover
5174010	LH Handle Hinge
5174186	Armboard Asm
5174245	Bracket
5174518	Link Bar Connector Pedal
5176295	Pin
5176314	Link Connector Undock
5176477	Shaft Connector Rocker
5176644	Link Down

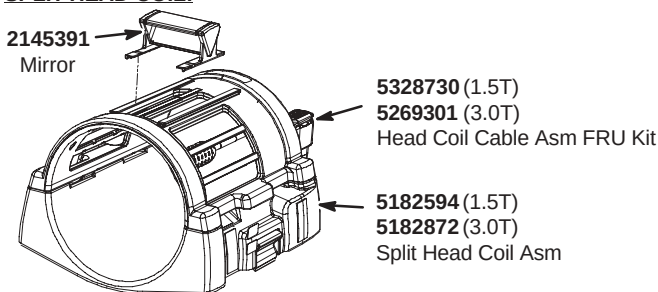
PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
5177611	Binding Head Slotted Screw, IN 10–32x0.5 SS 18–8
5177750	Pedal Stop Rubber
5178055	Connector Cover
5178055–2	Connector Cover
5178063	Shaft–linkage–right
5178101	Link, Left Front
5178144	Link, Right Front
5178145	Hexagonal Socket Head Screw, IN 8–32 X 3/8, Brass
5178145–2	Hexagonal Socket Head Screw, IN 8–32 X 1/2, Brass
5178210	Shaft–linkage–left
5178255	Link, left Rear
5178291	Cover Hinge
5178320	Connecting Rod
5178483	Link, Right Rear
5178611	Block–mounting
5178635	Extension–shaft
5178738	Pin–actuator
5178915	Link–reversing
5179251	Rear LH T & G Asm
5179251–2	Rear RH T & G Asm
5179376	Arm Board T & G Asm
5179376–2	Arm Board T & G Asm
5179600	Front LH T & G Asm
5179600–2	Front RH T & G Asm
5180380	Wheel Housing Asm
5181517	1.5T Table Cable Track
5181517–2	1.5T Table Cable Track
5181751	Reel
5183907	Decal Pedal Set
5184060	Lever
5189212	ODU Harness Cable
5189213	Flat Head Slotted Screw IN 1/4–20 x 3/4 Brass
5191326	Bushing
5193793	Harness Jumper Touch and GOC
5198459	Connector Return Spring
5198829	Shaft Spring Connection
5199333	Pedal Top Rubber
5199333–2	Pedal Top Rubber
5213295	Guide Strip
5213295–2	Guide Strip
5220240	Shoulder Screw, IN 1/4–20 X 3/8, SS18–8
5220240–2	Shoulder Screw, IN 1/4–20 X 5/8, SS 18–8

PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
5220240-3	Shoulder Screw, IN 1/4-20 X 1/2, SS 18-8
5220579	Nylon Washer
5220580	Rubber Washer
5263096	Bearing IGUS MCI-12-02
5263182	Rear Cradle Cover
5306407	Table Bumper FRU Kit
5338208	Wheel, Caster
5341331	Hydraulic Rotary Pump Asm
5341568	Hydraulic Rotary Pump Upgrade Kit
5344577	Service Filler Pad
1000904P41	0.266 ID, 0.75 OD, .064 THK Brass Washer
1000904P61	ID 0.391 OD 0.750 THK .031 Phosphor Bronze Washer
1000904P419	ID 0.328, OD 0.5, THK .016, Washer
1000904P425	0.2 ID, 0.5 OD, .04 THK Washer
1000904P460	0.201 ID, 0.375 OD, .05 Thick, Brass Washer
1000904P477	ID 0.531 OD 1.062 THK .095 ALUM 6061 Washer
1000904P478	ID 0.531 OD 1.062 THK .095 Brass
1000904P498	0.635 ID, 1~ OD, .125 Thick Aluminum Washer
1000904P502	0.406 ID, 1~ OD, .062 THK Nylon Washer
46-136342P23	Bearing Oil-imp Bronze, Sleeve, 0.365 ID, .503 OD. .313 Lg
46-136342P51	Bearing Oil-imp Bronze, Sleeve, 0.377 ID, .503 OD. .875 Lg
46-136342P60	Bearing Flange: OD 0.500~, THK .046~ OIL-IMP Bronze
46-136342P128	Bearing Flange: OD 0.625~, THK .3625~ OIL-IMP Bronze
46-136342P131	Bearing Oil-imp Bronze, Sleeve, 0.3155 ID, .4395 OD. .25 Lg
46-136353P105	Spring-inch Free LG 2.000 RATE LB/IN 2.1
46-136393P147	Groove Pin, .156 OD, .5 Lg
46-170024P1	Screw-inch 10-32 SLOT HD #303 SST
46-170024P12	Screw 8-32 SLOT HD #303 SST
46-170073P5	.002 Thk x 2.0 Wide x 60 Yd Roll
46-170490P38	Spring-inch Extension 2.5 Inches .25 Free Lgth 2.500 SST 316
46-170490P59	Spring-inch Extension 2.75 Inches .2 Free Lgth 2.75 SST
46-170490P60	Spring-inch Extension 4.5 Inches .5 Free Lgth 4.50 SST
46-170686P1	Loctite 242, 0.5CC Tube
46-208560P96	Screw N7715010 (Passivated)
46-208563P88	Hex Socket Set Screw SST 316, 10-32 x 5/16
46-208563P90	Hex Socket Set Screw SST 316, 8-32 x 3/16
46-208601P25	3/8~ Screw, Internal Tooth Phos Brz
46-208695P75	Heyco No.2210 SB-1375-16 Black
46-208695P89	Heyco Part No.2017 SB-375-281
46-208735P58	Split Lengthwise .187 Dia X 1.000 Lg .040 Wall 420 SST
46-208735P76	Pin, Spring, Dowel
46-208737P3	Hex Socket Head Cap Screw 6-32 x 3/8, SST

PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
46-208737P12	Hex Socket Head Cap Screw, 10-32 X .50, SST
46-208737P13	Hex Socket Head Cap Screw, 10-32 X 5/8, SST
46-208737P20	Hex Socket Head Cap Screw, 10-32 X 1.0, SST
46-208737P27	Hex Socket Head Cap Screw, 1/4-20 X 3/4, SST
46-208737P28	Hex Socket Head Cap Screw 10-32 X 3/4 SST
46-208737P47	Hex Socket Head Cap Screw, 3/8-16 x 3/4, SST
46-208746P9	Cable Tie Panduit TM3510C
46-208747P2	Adhesive Mount – Cable Tie
46-208758P1	3.62~ X .094~ Cable Tie
46-208758P5	Self Locking Cable Tie
46-208910P4	Screw 18-8 SST (Passivated)
46-208913P9	Brass Slotted Flt Hd Scr 10-32x3/4
46-208913P24	Brass Slotted Flt Hd Scr 1/4-20x1/2
46-208932P7	Hex Head Cap Screw, 3/8-16 x 7/8, SST Passivated
46-208932P10	Hex Head Cap Screw, 1/4-20 X .50, SST Passivated
46-208932P16	Hex Head Cap Screw, 1/4-20 X 2.0, SST Passivated
46-208935P5	6-32, .313 Hex X .109 Thk, Brass
46-208938P5	10-32, 3/8~ Hex X .130 Thk, 18-8 SST
46-208945P3	1/4-28, .438 Hex X .219 Thk, Brass
46-208946P4	Hexagon Type 18-8 SST Jam Nut
46-220184P18	Round Head Slotted Brass Screw, 6-32 x 1 Lg
46-220184P26	Round Head Slotted Brass Screw, 8-32 x 1/2 Lg
46-220196P3	Washer, #10 Screw Size (0.190), SST
46-220270P9	E-Clip 0.750 Shaft Dia, BE-CU
46-220277P18	E-Clip 0.500 Shaft Dia, BE-CU
46-220277P19	E-Clip 0.250 Shaft Dia, BE-CU
46-220277P22	E-Clip 0.312 Shaft Dia, BE-CU
46-220277P23	E-Clip 0.375 Shaft Dia, BE-CU
46-220277P24	E-Clip 0.625 Shaft Dia, BE-CU
46-220277P27	E-Clip 0.188 Shaft Dia, BE-CU
46-221304P7	0.188 DIA X 0.65 Length, 303 SST
46-221560P1	Hex Socket Oval Point Set Screw 1/4-20 x 3/4, SST Passivated
46-221560P3	Hex Socket Oval Point Set Screw 8-32 x 3/8, SST Passivated
46-221560P4	Hex Socket Oval Point Set Screw 10-32 x 1/2, SST Passivated
46-221806P2	Pan Head Phillips Screw, 10-32 x 1/2 SST Passivated
46-221922P7	Hex Head Screw, 10-32 x 1-1/4 Lg SST Passivated
46-230006P2	Hydraulic Lubricating Oil Red Colored Oil, 32oz.
46-230434P2	0.047 Wire Dia X 0.594 Lg 302 SST Hairpin Type
46-230947P1	.312~ I.D., .5~ O.D., .093~ Wall. tygon F-4040-A Tubing
46-230957P1	Loctite 680, 50ml Bottle
46-243178P3	Bearing, ID 0.500~, OD 1.375~, Width 0.375~ Open, Turcite Plastic

PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
46-244996P1	Bushing, Dock Guide Roller, 1~ Dia. Nylatron
46-256047P1	Mobiltemp Shc 32 14 Oz Cartridge
46-256049P16	Hole Dia 1.000, Head Dia 1.2187 O~all Ht .4062 Nylon White
46-256080P1	Cable,Push Pull,Nylon,Molybdenum Disulfite Impregnated
46-256081P6	3/8~ Screw Size, HI-collar
46-260789P88	Screw,.375-16 X 1.25 Soc Hd Capscrew 304 Stainless Steel
46-256193P9	3/8-24 Hex 0.564 Ht .453 Ring Type Nylon Locking Element
46-260797P82	Screw Binding Head
46-265887P2	Upper Scissors Roller Shaft, 3/8 Dia
46-265887P6	Lower Scissors Roller Shaft, 3/8 Dia
46-265888P1	Roller, 2 OD, .50 ID, .31 Lg
46-265892P1	Dock Hook, 3/8-24 Thd, SST, Hardened
46-265895P1	Pin, .187 OD, 4.23 Lg SST
46-265896P1	Brass Bushing, .837 ID
46-265897P1	Link Lever, Polycarbonate Plastic
46-265898P1	Link Lever, Polycarbonate Plastic
46-265899P1	6.7~ Long, .081 SST Spring Wire
46-265905P1	Foot Pedal Pad, Rubber
46-265906P1	Docking Guide Roller Shaft
46-265907P1	Casting
46-265908P1	Shaft, .625 OD, 6.56 Lg SST
46-271028P1	Hydraulic Cylinder To Table Top
46-271127P1	Molded Plastic W/Cap & Rubber Gasket
46-271129P1	Molded Polycarbonate
46-271134P1	1~ & .562~OD, .316~ID X.5~, Brass
46-271161P1	.375~& .312~ DIA, 1.17~LG, SST
46-271162P1	2.12~X1.19~, .5~ Thk Alum Bar
46-271178P1	Secondary Stop Pin, Brass
46-271179P1	Secondary Stop Pin Housing, Delrin
46-271192P1	.312~DIA, 6.45~LG, .125 Keyway, SST
46-271194P1	.312~DIA 6.88~ Long, SST
46-271195P2	Shaft, Hook Stop, .312 OD, 7.75 Lg, SST
46-271195P4	Shaft, Link Pivot, .312 OD, 0.89 Lg \SST
46-271195P7	Shaft, Dock Pedal, .312OD, 1.38 Lg SST
46-271195P8	Shaft, Down, .312 OD, 1.46 Lg SST
46-271195P11	Shaft, Dock Hook Pivot, .312 OD, 2.97 Lg SST
46-271232P1	Slide, White Delrin
46-271235P2	Key, 3/16~ SQ, .44 Long, Brass
46-271235P3	Key, 3/16~ SQ, .62 Long, Brass
46-271408P1	I.V. Bushing, 1.38~OD, .635~ Hole, 2.5~LG, Alum
46-271438P1	1~OD, .66~ID, White Velcro Hooks
46-271510P1	Shaft For Pedals

PATIENT TABLE PARTS <i>(NOTE: Patient table parts are listed in numerical order)</i>	
46-271511P1	Cable, 71~ (Trim Ar); Swaged Ball One End
46-271512P1	Adjustment Screw, 1/4-28 x 1.38 Lg
46-271551G1	Detachable, Telescoping IV Support
46-271554P1	Roller Wear Strip, 1.93 x 8.63 SST
46-271555P1	BNC Right Angle Adaptor
46-271562P1	Spacer, .39 ID, 1.00 OD, .40 Lg Nylon
46-271576G1	Mechanical Linkage
46-271587P1	2.19~ X 1.5~ Gusset, .081 Alum
46-271913P1	Bell Crank, Brass
46-271914P1	Pivot Stud, 1/4-28 Thread, .194 Hole, Brass
46-271946P1	Oil Filter, Reservoir Internal Cap Poly Foam
46-271966P1	Cylinder Mounting Pin .437~OD X 4.34
46-282216P1	Cable Clevis, 1~X.44~X.38~ .194 Hole, Brass
46-282440P1	Bushing, .442~OD .314~ID, .25~Thk, Bronze
46-307077P1	Flat Head Solid Rivet, 3/16 DIA X 3/8 3/8 Dia Head 18-8 SST
46-307878G1	Hydraulic Oil Filter And Gasket
46-317179P1	Squeeze-shut Ratchet-latch, 1/2~ Dia

SPLIT HEAD COIL:**Note: The following parts are not shown:**

5263175 (3.0T) QHTR Circuit Board Asm Kit

5328732 (1.5T) QHTR Circuit Board Asm Kit

5267144 Head Coil Cushion – Flat

5271434 Head Coil Cushion – Head

2355094 Replacement Spring Contacts Kit

PATIENT HANDLING FRU KIT's:

46-306574P7 0.5 x 9 x 14.5 Gray Rectangular Pad, Nylon Covered

46-306574P8 1.0 x 9 x 14.5 Gray Rectangular Pad, Nylon Covered

46-306574P9 0.5 x 9 x 36 Gray Rectangular Pad, Nylon Covered

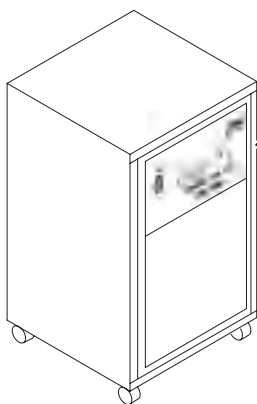
46-306574P10 1.0 x 9 x 36 Gray Rectangular Pad, Nylon Covered

46-317758P1 Pneumatic Patient Alert System

5122500 ECG Lead Kit

[illegible]

SHIELD COOLER



2188440-2
Sumitomo CSW-71D
Shield Cooler Compressor

OR

5314887
Sumitomo F-50
Shield Cooler Compressor

SHIELD COOLER FRU's Not Shown:

2154502-2 Run 621, Supply Gas Line, 65.6 ft or 20M long

2154505-2 Run 622, Return Gas Line, 65.6 ft or 20M long

2155316-2 Run623, Compressor to Pen Panel Cable, 49.21 ft or 15M long

2218192 Run624, Cold Head Control Cable

2214062 Run 832, LCC Sleeve and Cold Head Cable

2259837 LCC Cold Head Control Cable Asm

2200835 Power Cable for Sumitomo CSW-71D Shield Cooler Compressor

2211446-3 Compressor Cable Kit consists of:

2154502-2, Run 621

2154505-2, Run 622

2259837, LCC Cold Head Control Cable

2200835, Shield Cooler Compressor Power Cable

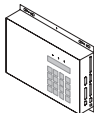
5335002 Flow Control Adapter, used with F-50 Compressor only

MAGNET RUNDOWN UNIT (MRU) FRU'S:

5196918	MAGNET RUNDOWN UNIT (Primary)
5196919	MAGNET RUNDOWN UNIT (Remote)
5363631	SPARE FUSE KIT includes: 1.25A, 250V Fuse; 2.5A, 250V Fuse; 400ma, 120V Fuse; 200ma, 250V fuse
5199490	BATTERY PACK ASM
5180324	ADAPTER PLATE
5196921	REMOTE TO PEN PANEL CABLE
5196920	PRIMARY TO PEN PANEL CABLE
2128699	MAGNET TO MRU CABLE, RUN 606

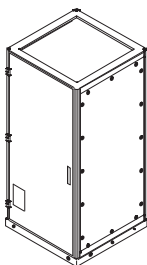
OXYGEN MONITOR (O2) FRU'S:

5390676	OXYGEN MONITOR FRU – Contains all of the parts included with the O2 monitor, box, sensor and cables
5399042	OXYGEN MONITOR CALIBRATION KIT – Contains all of the parts needed to calibrate either the nw or old O2 monitor
5390678-3	REMOTE 5-YEAR SENSOR – Sensor for the ISA-60M O2 monitor

MAGNET MONITOR:

2394952
MAGMON3 (MON)

HEC CABINET

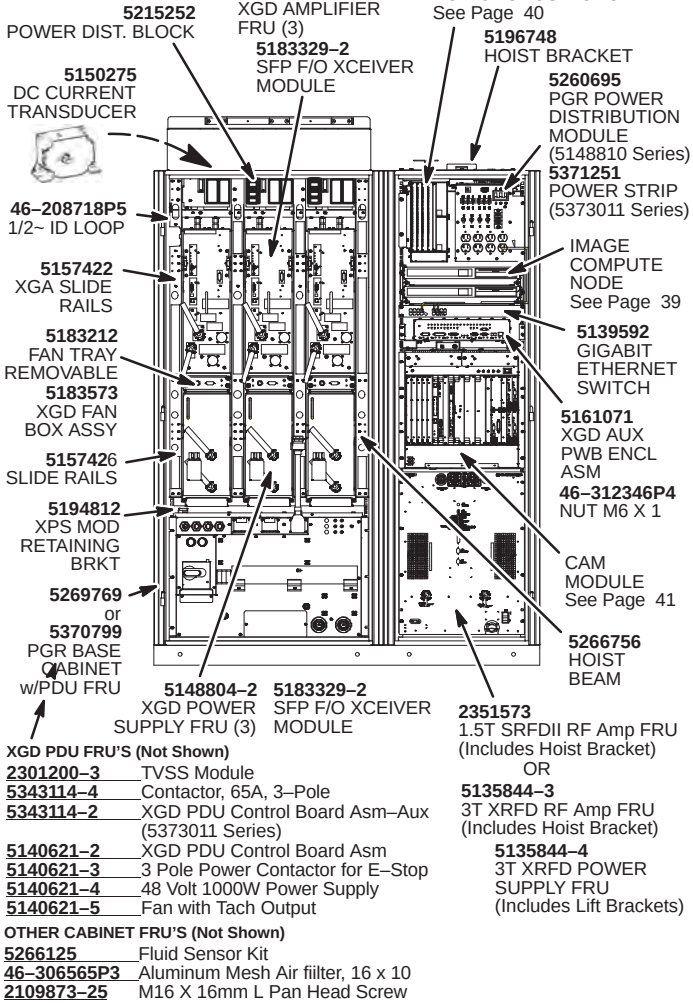


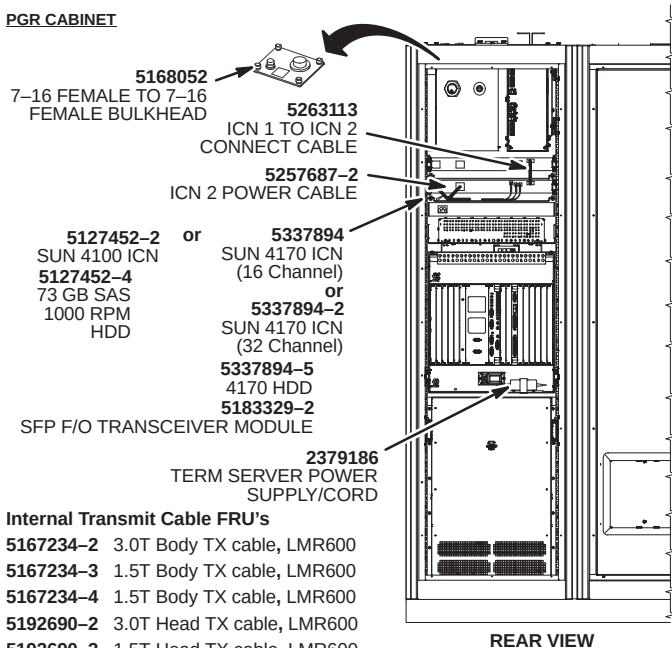
HEC (Heat Exchanger Cabinet) FRU KIT's

5264701	Deionization Filter Cartridges Kit
5156315	3hp Blower Kit
5156316	Cryogen Compressor Flow Meter Kit
5156318	Mixing Valve Kit for Gradient Coil and PGR Cooling Loops
5156319	Power Box Kit
5156320	Mixing Valve Kit for Chilled Air Blower
5156321	5hp Pump Kit
5262029	Equipment Room Hoist Support Tube
5266827	Signal Box Fuse Kit
5267643	Liquid Pressure Sensor Kit
5267644	Level Switch Kit
5267648	Pressure Difference Sensor Signal Box Circuit Board Kit
5267649	Resistivity Sensor Kit
5268665	Liquid Temperature Sensor Kit
5269683	Coolant Removal Kit
5269685	Cryogen Compressor Filter Kit
5269689	Air Temperature Sensor Kit
5269691	Signal Box Kit
5357155	Cryo Compressor Flare Fitting FRU Kit
5376555	Cryo Supply Header
5400660	Facility Tee with QD

HEC COOLING SYSTEM FRU's

5267596	Liquid Cooling Valve Kit (Contains ball valves and check valves)
5174313-4	Gradient Coil Coolant, 4 Gallons – non-copper corrosive, algae resistant

PGR CABINET

PGR CABINET**Internal Transmit Cable FRU's**

- 5167234-2** 3.0T Body TX cable, LMR600
5167234-3 1.5T Body TX cable, LMR600
5167234-4 1.5T Body TX cable, LMR600
5192690-2 3.0T Head TX cable, LMR600
5192690-3 1.5T Head TX cable, LMR600
5192690-4 1.5T Head TX cable, LMR600

5304065 (5148810 Series) & 5388782 (5373011 Series)**Internal Signal Cable FRU Kit contains:**

5178931	5165870	5165970	5176566	5182238	5182238-2
5182238-3	5182238-4	5184303	5257682	5257683	5257683-2
5257684	5257685	5257685-2	5257685-3	5257685-4	5257685-5
5257685-6	5257685-10	5167344	5167005	5257680	5257681
5166354	5166222-13	5166222-18	5257685-11	5257685-12	5257685-13
5176990	5176262	5176262-2	5176262-3		
5174739 (5148810 Series Only)			5376191 (5373011 Series Only)		

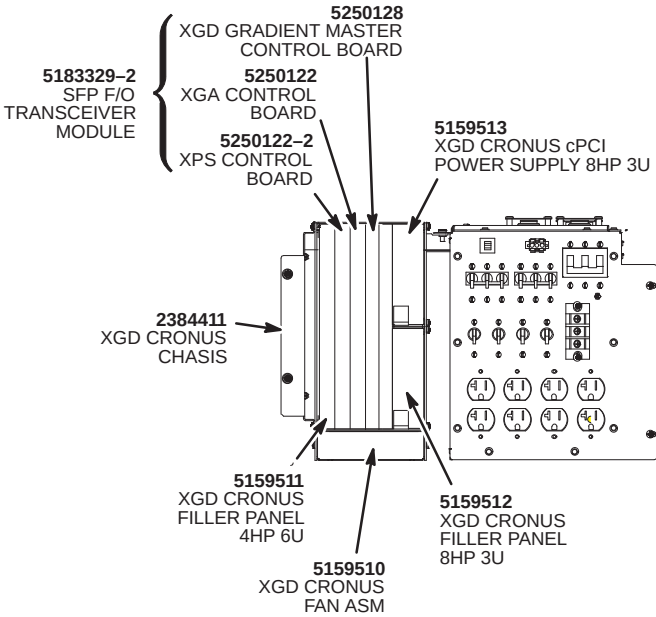
5304386, Internal Fiber Optic Cable FRU Kit contains:

5182888	5182317	5182317-2	5182317-3	5182317-4	5182317-5
5182317-6	5257686				

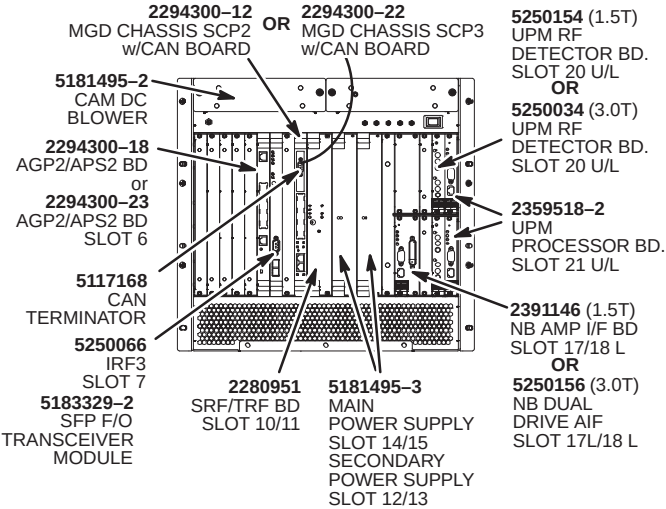
5304635, Internal Power Cable FRU Kit contains:

5173399	5164904	5165800	5172391	5174242	5174242-2
5183977					

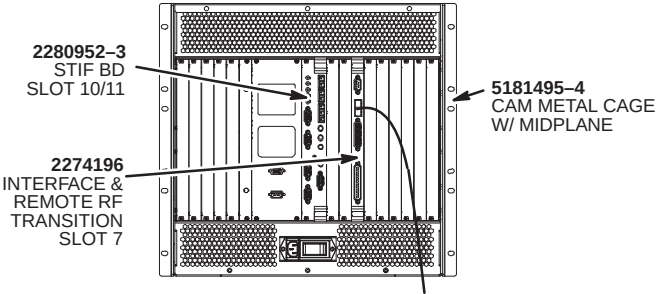
CRONUS MODULE

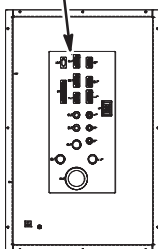
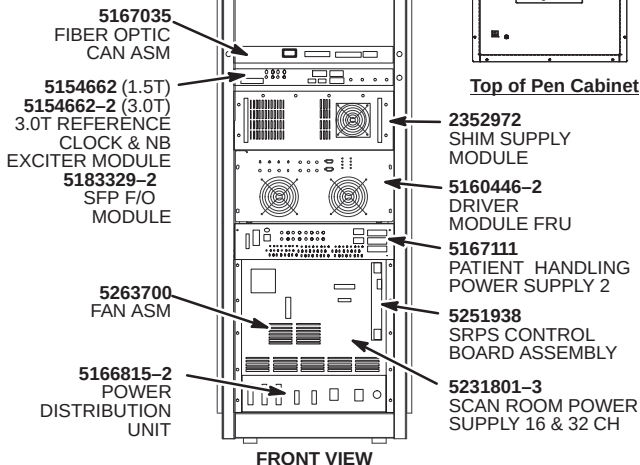


CAM MODULE (FRONT)



CAM MODULE (REAR)



PEN CABINET**5199042
TOP IO PANEL****Top of Pen Cabinet****INTERNAL CABLE FRU KIT'S****5308001-2, Power Cable FRU Kit contains:**

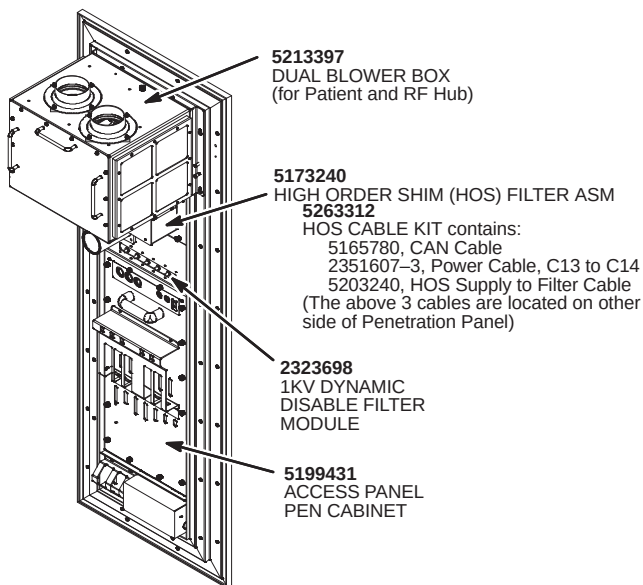
5167143	5167143-2	5172838	5155030	5172849	5171737
5154994					

5308001-3, RF Cable FRU Kit contains:

5181464	5166512-3	5166512-2	5172955-2	5172955	5166749
5166749-9	5172679-10	5172679-11	5172679		

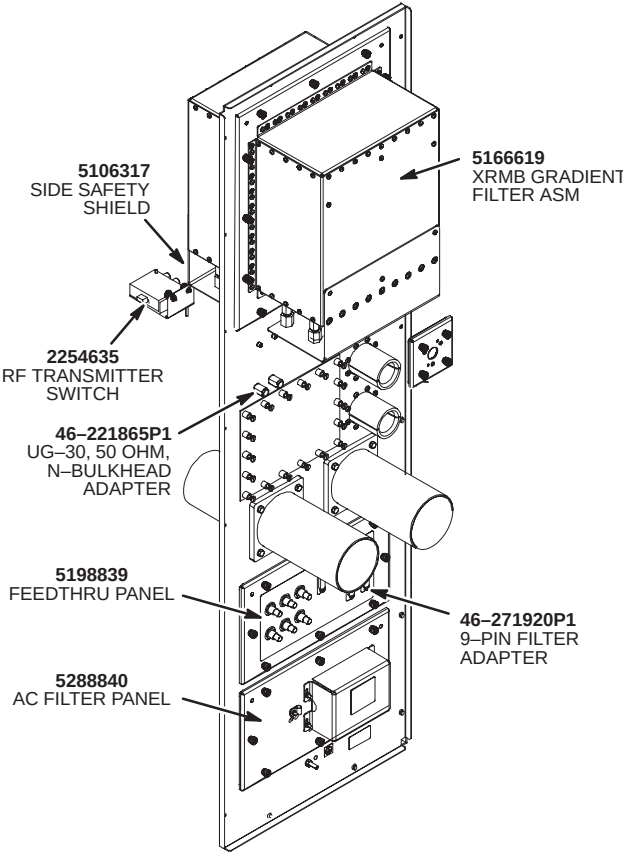
5308001-4, Power Distribution Cable FRU Kit contains:

2322434	5168641	5168043	5168317	5167882	5166461
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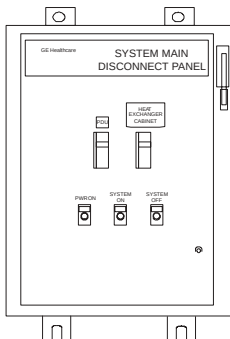
PEN CABINET PENTRATION PANEL**REAR VIEW****INTERNAL CABLE FRU KIT'S (Continued from previous page)****5308001-5, Signal Cable FRU Kit contains:**

5165866	5166997	5169443	5172726	5166354-2	5171626
5170521	5172868	5167365	5171583	5172959	5169381
5168461	5168597	5172812	5172595-2	5169345	5175777
5172595					

SPW (Secondary PEN Wall)

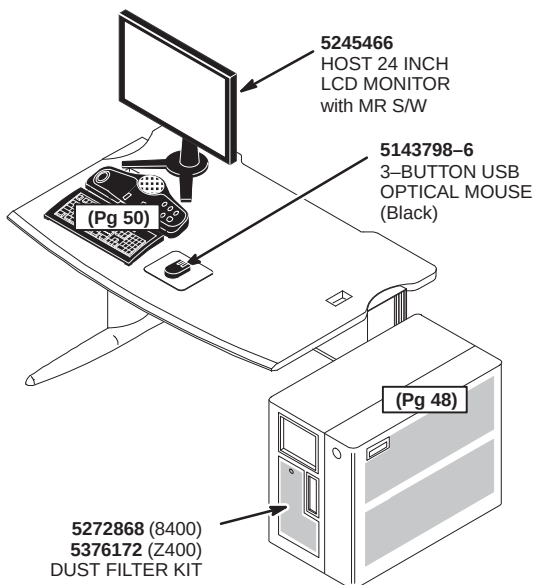


[illegible]

MAIN DISCONNECT PANEL (MDP)**MDP FRU's Not Shown**

<u>5161287-2</u>	1.5A, 380/480V Fuse
<u>5161287-3</u>	60A, 380/480V Fuse
<u>5161287-4</u>	Green PB Lens
<u>5161287-5</u>	Green PB LED 120V Power Supply
<u>2375862-2</u>	200A, 600V Fuse
<u>2276547-14</u>	12V Battery
<u>2276547-16</u>	12V Battery Charger
<u>5390238</u>	Battery
<u>5390239</u>	Power Supply
<u>5376545</u>	Battery and Charger Kit

OPERATOR WORKSPACE



OPERATOR WORKSPACE FRU's Not Shown

2156158-63 1.5T Software Collector for 20.1 Release

2156158-74 3.0T Software Collector for 20.x Release M4

2156158-80 DV Apps Software Collector (22.x only)

5165713 DVD Media for Linux

GOC COMPUTER CABINET WITH HP8400;**5267342**

GOC CableCollector
(See details below)

5133768-2
FIBER OPTIC
CONVERTER
BOX

5139480-2
8-PORIT SWITCH

5262750
PDU ASM

2395043 or 2395043-2
GOC AUDIO ASM

5183547-30

HOSTPC

5183547-15

73GB SAS HARD
DRIVE FOR PC

5183547-20

SINGLE 1GB DIMM

5183547-36

SINGLE 2GB DIMM

5183547-22

DVD-ROM DRIVE

5184500

USB to RS422
CONVERTER
(Connected to
PC USB port)

46-271382P6
1 POLE, 15 AMP
CIRCUIT BREAKER

5265358

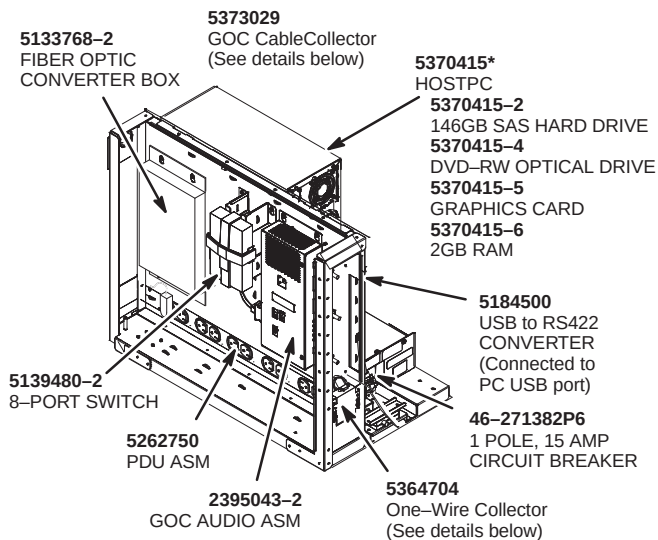
One-Wire Collector
(See details below)

5267342 GOC Cable Collector FRU consists of;

- 5270774** Host Power Cable
- 5270803** Main LCD Monitor Power Cable
- 5263413** Run E3046 Main Monitor DVI-D Cable
- 5264533** Run E3047 SCIM-GOCAA Cable
- 5267826** Audio Cable Asm
- 5270723** GOCAA-PDU DC Power Cable
- 5267389** Run E3045 Modem Cable
- 5272839** Host Hub Ethernet Cable
- 5272839-2** Hub-Front Port Ethernet Cable
- 5264521** Run E3048 USB Extension Cable
- 5169105** Run E3016 Y6 Cable
- 5265059** Host to IRDC DVI-D Cable
- 5265784** Host to IRDC USB Cable
- 5270246** IRDC Power Cable

5265358 One-Wire Collector FRU consists of:

Humidity & Temperature iButton, iButton Holder, USB to RS422 Converter, USB and Sensor cables, Run E3019, and Dual Locking Tape

GOC COMPUTER CABINET WITH HP Z400;

5373029 GOC Cable Collector FRU consists of;

5330291 Run E3046 Main Monitor DVI-D Cable

5330293 Run E3047 SCIM-GOCAA Cable

5330292 Run E3048 USB Extension Cable

5330289 Run E3016 Y6 Cable

5330294 Host to IRDC DVI-D Cable

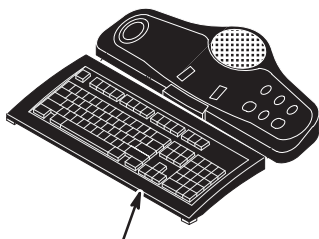
5330295 Host to IRDC USB Cable

5330303 IRDC Power Cable

5364704 One-Wire Collector FRU consists of:

Humidity & Temperature iButton, iButton Holder, USB to RS422 Converter, USB and Sensor cables, Run E3019, and Dual Locking Tape

* The 5370415 Z400 Host computer only compatible with **2156158-80**, DV Apps software collector

SCIM & KEYBOARD:**NOTE:**

If Keyboard is to be replaced, also order **46-317475P2**, 2 inch x 4 inch adhesive backed velcro to attach to bottom of new keyboard.

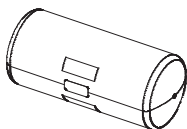
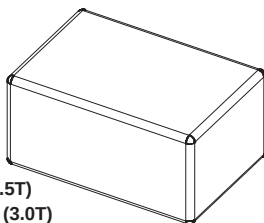
KEYBOARDS:

- 5143798-7** Standard USB Keyboard – English
- 5143798-8** Standard USB Keyboard – French
- 5143798-9** Standard USB Keyboard – German
- 5143798-10** Standard USB Keyboard – Spanish
- 5143798-11** Standard USB Keyboard – Italian
- 5143798-12** Standard USB Keyboard – Portuguese
- 5143798-13** Standard USB Keyboard – Scandinavian
- 5183547-46** Standard USB Keyboard – Danish
- 5183547-47** Standard USB Keyboard – Dutch
- 5183547-48** Standard USB Keyboard – Norwegian

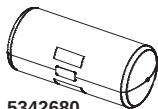
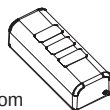
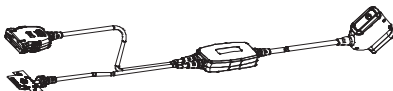
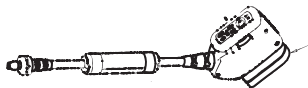
SCIM OVERLAYS:

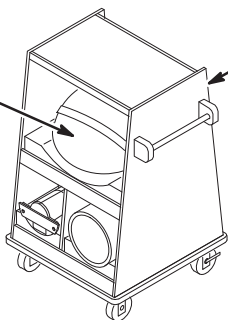
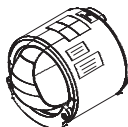
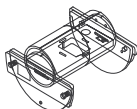
- 5148389-2** MR SCIM – Black with English Overlay
- 5148389-9** MR SCIM – Black Coordinated Global Overlay Set

[illegible]

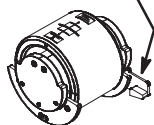
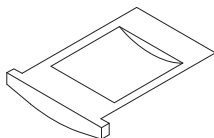
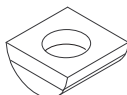
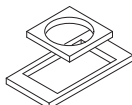
UNIFIED SURFACE COIL PHANTOM SET**5342679 (1.5T)****5342679-2 (3.0T)**Large Cylindrical
Unified Phantom**5343347 (1.5T)****5343347-2 (3.0T)**

TL Unified Phantom

**5342680**Small Cylindrical
Unified Phantom**5342681**Cubical
Unified Phantom**COIL CABLE FRU'S****2424308**1.5T 12-Channel BA
Output Staging Cable
(P-Connector)**2424269**1.5T 16-Channel HNS
System Staging Cable
(P-Connector)**SERVICE KITS****CABLE FRU KIT's****5306523**MR450/MR750 Service Cable Hardware Kit for Troubleshooting
(Includes BNC components, studs, loopbacks, gender changers)

1.5T SERVICE PHANTOMS and ACCESSORIES**2274476**LV SHIM PHANTOM
INNER SPHERE**2306642**LV PHANTOM
SMALL HALF ASM**2306645**LV PHANTOM
BIG HALF ASM**2134213**SPT PHANTOM
CART**2125244**SPT SHORT LOADER
w/SPHERE ASM**2131027-2**

DQA III ASM

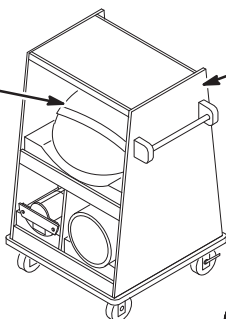
5121115LOCATION
BRACKET
w/SCREWS**2321556**DQA PHANTOM
WITH LOADER**5110241**HEAD LOADER
POSITIONER**2291292**EPI HEAD
POSITIONER**2170481**EPI FOAM
POSITIONER**46-317586G1**10cm NiCl
PHANTOM**2424387**TORSO COIL
PHANTOM POSITIONER**2258366**10cm SILICON
FILLED PHANTOM**SERVICE COILS****53**

3.0T SERVICE PHANTOMS and ACCESSORIES

2360051
LV SHIM PHANTOM
INNER SPERE

2360049
LV PHANTOM
SMALL HALF ASM

2360050
LV PHANTOM
BIG HALF ASM



2134213
SPT PHANTOM
CART



2359877
HEAD
SPHERE

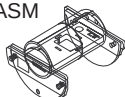


2360034
100mm
SPHERE

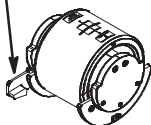


2152220
MNS PHANTOM

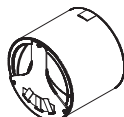
2131027-2
DQA III ASM



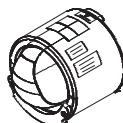
5121115
LOCATION
BRACKET
w/SCREWS



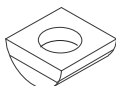
2371512
DQA PHANTOM
WITH LOADER



2371511
BODY LOADER
w/SPHERE ASM
For Field Replacement Order:
2360025– BODY SPHERE
2360037 – BODY LOADER



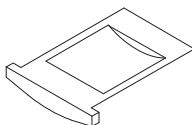
2360031
HEAD LOADER



2291292
EPI HEAD
POSITIONER



2170481
EPI FOAM
POSITIONER



5110241
HEAD LOADER
POSITIONER

CABLE COLLECTOR FRU'S**5192499, 3.0T Rear Pedestal Cable FRU Kit contains:**

Run M3320 Run M3321 Run M1315 Run M1335 Run M2300 Run M3351
Run M1324

5308827, 1.5T Rear Pedestal Cable FRU Kit contains:

Run M3320 Run M3321 Run M1315 Run M1335 Run M2300 Run M3351

5192500, Rear Pedestal 32-Channel Add-on Cable FRU Kit contains:

Run M3322 Run M3323 Run M3330 Run M3331 Run M3332 Run M3333
Run M3371 Run M3372 Run M3373

5331721, Scan Room Base SHORT Cable FRU Kit contains:

Run M1302 Run M1303 Run M1304 Run M1310 Run M1311 Run M1312
Run M2025 Run M3030 Run M3031 Run M3300 Run M3301 Run M3302
Run M3303 Run M3304 Run M3306 Run M3307 Run M3308 Run M3309
Run M3310 Run M3311 Run M3312 Run M3313 Run M3315 Run M3316
Run M4005 13W6 Shunt Connector for 16 & 32 Channel Only

5331721-2, Scan Room Base LONG Cable FRU Kit contains:

Run M1302 Run M1303 Run M1304 Run M1310 Run M1311 Run M1312
Run M2025 Run M3030 Run M3031 Run M3300 Run M3301 Run M3302
Run M3303 Run M3304 Run M3306 Run M3307 Run M3308 Run M3309
Run M3310 Run M3311 Run M3312 Run M3313 Run M3315 Run M3316
Run M4005 13W6 Shunt Connector for 16 & 32 Channel Only

5191278, 3.0T Scan Room Transmit SHORT Cable FRU Kit contains:

Run M1305 Run M1306 Run M1001 Run M1002 Run M1006 Run M1015

5191278-2, 3.0T Scan Room Transmit LONG Cable FRU Kit contains:

Run M1305 Run M1306 Run M1001 Run M1002 Run M1006 Run M1015

5191368, 1.5T Scan Room Transmit SHORT Cable FRU Kit contains:

Run M1305 Run M1306 Run M1001 Run M1002 Run M1341 Run M1342

5191368-2, 1.5T Scan Room Transmit LONG Cable FRU Kit contains:

Run M1305 Run M1306 Run M1001 Run M1002 Run M1341 Run M1342

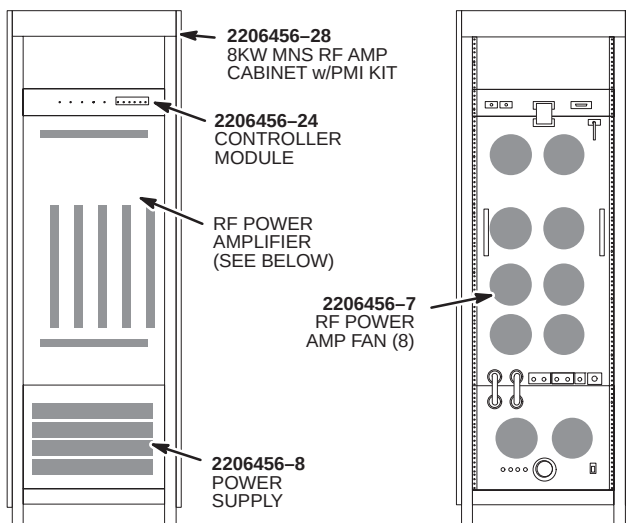
5343119, Fiber Optic Cable FRU Kit contains:

Run P2003 Run P2004

5343120, In-Room Display (IRD) Cable FRU Kit contains:

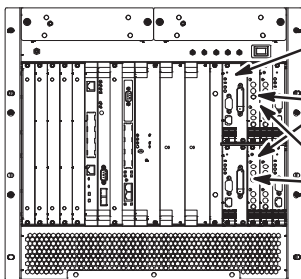
Run P2026 Run M3366

5169566 Run 3367, MAG-TDM-J1 to MAG-RX1, 44-pin HD sub-D 560mm
5169566-2 Run 3368, MAG-TDM-J2 to MAG-RX2, 44-pin HD sub-D 1360mm
5180907 Run 3350, MAG-SRI-J10 to MAG-BORE, Bore Temp, 3000mm

8KW MNS CABINET

NOTE: The following RF POWER AMP internal parts are Not Shown:

- 2206456-2** RF Power Amp Preamp/Splitter
- 2206456-3** RF Amplifier (5)
- 2206456-4** RF Power Amp Combiner/Coupler
- 2206456-5** RF Power Amp Coupler (4)
- 2206456-6** RF Power Amp Bias Tee
- 2206456-13** Cable – Splitter To Amplifier (5)
- 2206456-20** Coupler–bias Tee
- 2206456-21** Power Bias Tee
- 2206456-22** Option 10, PMI Interface Kit
- 2206456-23** Option 11, PMI Interface Kit
- 2206456-25** Option 12, PMI Interface Kit
- 46-170021P29** 5 Amp Fuses (3)

MNS COMPONENTS

5250032
8kW BB Amp I/F BD.
Slots 17/18 U

5250034
RF DETECTOR BD.
Slots 19 U/L

2329754
50 OHM SMB
TERMINATOR

46-208990P1 50 Ohm BNC Bulkhead Adaptor

5300204-2 MR750 3.0T MNS Exciter (located in PEN Cabinet)

5189655-2 MNS Upconverter Module (located in Rear Ped)

2109880-2 Plain Washer, 6.4mm x 12mm

5263595, MNS Internal Cable FRU Kit contains:

5166222-14	5166222-15	5166222-16	5166222-17	5271096	5172955-3
5271821	5265145	5262319	5167882-2	5266021	5165866-2

5263595, MNS Interconnect Cable FRU Kit contains:

5265483	MNS Gigalink Cable
5169804-15	Run E1007
5169804-16	Run E1008
5169804-17	Run E1009
5169804-18	Run E1010
5169804-19	Run E1012
5169804-31	Run E1309
5169227-4	Run E3010
5270425	MNS Amplifier BNC Cable
5265976	MNS BB Exciter Power
5166749-8	Run M1308
5181464-2	MNS Cable
5181464-3	Run M1353
5179345	Run M3309
5173115-6	Run M1307
5181541	Run M3356
5191729-3	MNS 8w8 Cable
5271444	3T Cable Assembly MNS Adaptor
5310317	Run E0006

MNS COMPONENTS

2354050-6 3T 13C TR Switch (located on LPCA)

2109880-2 Lock Washer, 4.1mm x 7.6mm (6)

2109873-15 Pan Head Screw, M4 x 12mm (6)

2109878-2 Plain Washer, 4.3mm x 9mm (6)

5114799-2 3T 13C MNS Preamplifier

2359525 3T MNS QD Box Linear Coils

2359525-2 3T MNS QD Box Quadrature Coils

2379500 2kW MNS QD Box with One BNC

2379500-2 2kW MNS QD Box with Two BNC

5114403-3 MNS QA Probe 3T 13C 32.125 MHz

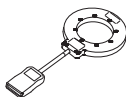
5114978-4 MNS QA Probe 31P

5114978-3 13C Sample Vial



2152220

MNS PHANTOM



2386800

MNS SERVICE COIL



2389665

MNS PHANTOM
POSITIONER

[illegible]

REVISION HISTORY

REV/DATE REASON FOR CHANGE

- | | | |
|---|------------|---|
| 1 | Jun 27, 08 | Initial release |
| 2 | Oct 11, 08 | Reformatted FRU manual into pocket size and added illustrations to help identify parts. |
| 3 | May 15, 09 | Added MR450 to title to reflect addition of MR450 1.5T components to manual. Update PAC part # and add PPG cable (ECO 2069724). Added cable FRU's (PQR 13250844). Updated XRMB FRU to 2144093-12 (PQR13223405). Added OW HP8400 FRU's (PCN's 2038518 & 2059074) |
| 4 | Feb 3, 10 | Added Preamp & Combiner FRU 2377703 (ECO 2062777). Updated RF Hub Backplane to 5159474-32 (ECO 2078155). Updated ODU Dock cable to 5328862 (ECO 2083025). Added new FRU's (ECO 2053514, 2096463). Added Dock Hook, Extension Spring, E-Clip, Hydraulic Reservoir Sponge, and Uplimit Cable to Patient Table FRU's (PQR 13278899 & 13280426). Added Sumitomo F-50 Compressor & flow control adapter (ECO 2059723). Corrected Cronus module board name to XGA (MRIhc46702). Corrected SPW FRU's (MRIhc46065). Added new APS2 bd, 2294300-23 (ECO 2064898). Corrected fuse part number in MDP (MRIhc46829). Added new GOCAA 2395043-2 (ECO 2084000). Added 1.5T Unified Surface Coil Phantom Set (ECO 2081133). Updated Scan Room short and long cable FRU kit part #'s (ECO 2089895). |
| 5 | Aug 2, 10 | Added new top laser mounting bracket for IRD (MRIhc49551). Added 5357155, 5376555 & 5400660 to HEC (MRIhcc49551). Added new 4170 ICN's to PGR (ECO 2090162). Revised OW monitor part to 5245466 (MRIhc49551). Added O2 monitor FRU's (2098349). Added MRU FRU's (ECR 2080462). Added SFP fiber optic transceiver module to PGR section (MRIhc5145). Added expanded FRU parts to Patient Table (ECR 2102683). Added 9-Pin filter adapter to SPW (MRIhc51545). Added HP Z400 computer (ECO 2056587). Added MDP battery and charger FRU's (ECO 2097255). Revised SRPS FRU to 5231801-3 (CAPA 845676). Added Unified Phantom set (MRIhc49551). |